

Enes Senel

Senior Scientist, Computational Biology — single-cell multi-omics, machine learning, spatial omics

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github.com/enessenel · [Google Scholar](https://scholar.google.com/citations?user=...)

PROFILE

Computer scientist and computational biologist leading single-cell work in oncology and immunology at Johnson & Johnson. My experience spans spatial transcriptomics, multi-omics integration, computer vision and medical robotics. I develop computational methods and reproducible analysis pipelines, set analytical direction across projects, and work with laboratory, clinical and manufacturing colleagues. I mentor junior scientists and manage external consultants.

CORE EXPERTISE

Biology: single-cell RNA-seq, CITE-seq, spatial transcriptomics, biomarker discovery, oncology and immunology, cell therapy and CMC analytics.

Methods: graph neural networks, optimal transport, deep learning, multimodal integration, domain adaptation, protein language models and real-time computer vision.

Leadership: analytical strategy, methodology review, mentorship, consultant management and cross-functional scientific collaboration.

PROFESSIONAL EXPERIENCE — JOHNSON & JOHNSON

Senior Scientist, Computational Biology — Johnson & Johnson Innovative Medicine, Berlin, Germany

Aug 2025 — Present

- Lead single-cell multi-omics programs (transcriptomics, proteomics, spatial) that feed biomarker discovery, mechanism-of-action work, and translational strategy for oncology and immunology assets.
- Set the analytical direction across several parallel projects: defining standards, reviewing methodology, and agreeing deliverables with clinical and discovery stakeholders.
- Mentor junior scientists and manage external consultants.
- Work with wet-lab, clinical, and CMC teams to turn analyses into experimental designs and go/no-go decisions.

Postdoctoral Data Scientist, Computational Biology — Johnson & Johnson Innovative Medicine, Berlin, Germany

Oct 2023 — Jul 2025

- Built automated Python pipelines for scRNA-seq, CITE-seq, and multi-modal analyses, replacing manual workflows that previously took days per dataset.
- Led the single-cell analyses behind biomarker discovery for a clinical-stage cell therapy and process characterisation for CMC.
- Set up the team's reproducibility conventions — containerisation, version-controlled pipelines, shared QC standards — now the default for new projects.
- Reviewed colleagues' analyses as the team's main technical reviewer.

RESEARCH AND ENGINEERING EXPERIENCE

Doctoral Researcher — Computational Biology — Max Delbrück Center for Molecular Medicine, Berlin, Germany

Oct 2019 — Sep 2023

- Designed and led Optocoder, a machine-learning pipeline for decoding barcoded transcripts from imaging-based spatial transcriptomics. Published in NAR Genomics & Bioinformatics.
- Rewrote novoSpaRc (optimal-transport reconstruction of tissue architecture from scRNA-seq) for scale, making it usable on whole-tissue datasets. Co-authored the Nature Protocols paper.
- Applied graph neural networks to spatial omics data, modelling cells as graphs over tissue structure to capture neighbourhood effects that expression-only models miss.
- Co-inventor on a US patent for 3D spatial gene-expression reconstruction.
- Handled the full cycle for both tools: problem framing, method development, benchmarking, release, documentation, and user support.

Visiting Scientist — The Hebrew University of Jerusalem, Rehovot, Israel

Feb 2023 — Apr 2023

- Ran an independent short-term project applying protein language models (ProtBERT) to find candidate anti-insecticidal proteins from raw sequence data, combining transfer learning with biological priors to prioritise candidates for validation.

Machine Learning Research Engineer — Coriolis Pharma GmbH, Munich, Germany

Apr 2019 — Sep 2019

- Developed deep-learning models for automated particle detection and classification from flow-microscopy images used in biopharmaceutical QC.
- Packaged the whole thing — ingestion, training, evaluation, reporting — as a Python/PyTorch/TensorFlow tool that internal scientists could run themselves.

Research Assistant — Computational Neuroscience — Max Planck Institute for Brain Research, Frankfurt am Main, Germany

May 2018 — Sep 2019

- Applied supervised ML to activity-dependent changes in the neuronal proteome.
- Built a multimodal integration and domain, adaptation pipeline linking proteomic and transcriptomic datasets.

Undergraduate Research Assistant — Medical Robotics — Ozyegin University Robotics Lab, Istanbul, Turkey

2014 — 2016

- Developed real-time needle-tip localisation and tracking from ultrasound imaging for autonomous robotic control during biopsy.
- Wrote the C++/CUDA interface connecting real-time image analysis to robot control.

SCIENTIFIC SERVICE

Helmholtz Juniors — Member (2022-2023). Elected representative for approximately 8,000 doctoral students across the Helmholtz Association; contributed to PhD survey analysis and per-centre reports.

MDC PhD Representatives — Awareness team (2020-2023). Worked on doctoral contracts, salaries and working conditions.

TECHNICAL SKILLS AND LANGUAGES

Programming and tools: Python, C++, CUDA, Java, R, SQL, PyTorch, TensorFlow, Keras, scikit-learn, NumPy, scanpy, Nextflow, Jupyter, containerisation and version-controlled workflows.

Languages: Turkish (native), English (fluent), German (advanced).

SELECTED PROJECTS

Optocoder

Lead author | 2020–2022 | Python, NumPy, scikit-learn

Developed a computational pipeline to decode spatially indexed bead arrays from multi-cycle microscopy images. The workflow aligns imaging cycles, calls bases, corrects phasing and prephasing, and trains an ML model against Illumina ground truth where available. Produces bead-to-barcode tables for downstream spatial analysis. Open-source GPL-3.0 release with configuration files and documentation.

novoSpaRc

Co-first author | 2020–2021 | Python, Jupyter, optimal transport

Co-led the Nature Protocols release of a method for reconstructing tissue architecture from single-cell gene expression. Uses optimal transport to combine expression similarity with partial spatial reference information. Focused on scalability for larger tissue datasets and a tutorial-driven API. Code: github.com/rajewsky-lab/novosparc.

Protein language models for agricultural biology

Visiting scientist | February–April 2023 | ProtBERT, transfer learning

Built a pipeline to identify candidate anti-insecticidal proteins from amino-acid sequences. Combined transformer embeddings with biological domain knowledge and classification heads to prioritise candidates for wet-lab validation.

Flow-imaging deep learning

Coriolis Pharma | 2019 | Python, PyTorch, TensorFlow

Developed particle detection and classification for pharmaceutical quality control. Packaged image ingestion, training, evaluation and reporting so scientists could retrain and audit models. The collaboration with Leiden University resulted in a Cytotherapy publication.

Ultrasound needle tracking

Ozyegin Robotics Lab | 2014–2016 | C++, CUDA, OpenCV

Developed real-time needle-tip localisation from ultrasound for image-guided robotic biopsy. Connected image analysis to robot control through a C++/CUDA interface and contributed Gabor-filter-based detection methods and five robotics publications.

EDUCATION

PhD in Computational Biology — Humboldt University of Berlin / Max Delbrück Center

2019–2026 (expected)

- Thesis: Computational modelling and machine learning for spatial transcriptomics and cell-cell interactions.

MSc in Computer Science — Technical University of Munich

2016–2019

- Thesis: Analysis of activity-dependent proteomic changes associated with homeostatic scaling. DAAD MSc Scholarship.

BSc in Computer Science — Ozyegin University, Istanbul

2012–2016

- Thesis: A standalone application for a 2D ultrasound-guided medical robotic system.

PUBLICATIONS

Benchmarking single-cell foundation models for real-world RNA-seq data integration

S. Han, T. Sztanka-Toth, E. Senel, A. Elnaggar, J. Patel, T. Mansi, D. Smirnov, J. Greshock, A. Javidi
bioRxiv (2026) — Preprint

Benchmarking Next-Generation Sequencing Platforms: A Comprehensive Comparison of Single-Cell RNA-Seq from Ultima UG 100 vs. Illumina NovaSeq X Plus

R. Xiong, J. Patel, T. Sztanka-Toth, E. Senel, K. Calara-Nielsen, Y. Rajpurohit, T. Altenburg, J. Greshock, D. Smirnov, A. Javidi
bioRxiv (2026) — Preprint

Ciltacabtagene autoleucel biomarkers of response in relapsed/refractory multiple myeloma patients from CARTITUDE-1: a phase 1b/2 study

R.M. de Oca, F. Wang, M. Hao, D. Smirnov, H. Zhao, M. Gormley, E. Zudaire, S.S. Chang, B. Foulk, J. Patel, E. Senel, D. Wu, C. Davis, R. Boominathan, A. Javidi, J. Zhu, O. Lau, K. Clara-Nielsen, Y. Rajpurohit, D. Madduri, S. Jagannath, Y. Lin, A. Cohen, V. Plaks
Blood Cancer Discovery (2025) — Under revision

Optocoder: computational decoding of spatially indexed bead arrays

E. Senel, N. Rajewsky, N. Karaiskos
NAR Genomics and Bioinformatics (2022)

novoSpaRc: flexible spatial reconstruction of single-cell gene expression with optimal transport

N. Moriel*, E. Senel*, N. Friedman, N. Rajewsky, N. Karaiskos, M. Nitzan
Nature Protocols (2021)

Particulate impurities in cell-based medicinal products traced by flow imaging microscopy combined with deep learning for image analysis

A.D. Grabarek, E. Senel, T. Menzen, K.H. Hoogendoorn, K. Pike-Overet, A. Hawe, W. Jiskoot
Cytotherapy (2021)

Gabor filter-based localization of straight and curved needles in 2D ultrasound images

M. Kaya, E. Senel, O. Bebek
Turkish Journal of Electrical Engineering & Computer Sciences (2020)

Visual needle tip tracking in 2D US guided robotic interventions

M. Kaya, E. Senel, A. Ahmad, O. Bebek
Mechatronics (2019)

Visual tracking of multiple moving targets in 2D ultrasound guided robotic percutaneous interventions

M. Kaya, E. Senel, A. Ahmad, O. Bebek
ICRA, Singapore (2017)

Visual tracking of biopsy needles in 2D ultrasound images

M. Kaya, E. Senel, A. Ahmad, O. Bebek
ICRA, Stockholm (2016)

Real-time needle tip localization in 2D ultrasound images for robotic biopsies

A. Ahmad, O. Orhan, E. Senel, O. Bebek
ICAR, Istanbul (2015)

PATENT

Method and system for 3D reconstruction of tissue gene expression data. Co-inventor. Patent publication: US20240257914A1.